

Spacial Mass distributions and Galactic Stability

Advisor: Yuan Shi

- Aim
- Overview
- Status

Aim of my research

- Dark Matter has shown little experimental success.
- Can a universal non-constant mass distribution remedy the apparent motion?

Project Overview

- Warm up: find lagrangian associated with $z(t) = \frac{\sqrt{a^2 t^2 + 1} - 1}{a}$
- Warm up: by proxy mass distribution $\mathcal{L} = -m(z)\sqrt{1 - \dot{z}^2}$
- Main: Extend this to 2-D space
- Main: Compare to data

- I have found $m(z) = \frac{m_0}{az+1}$ and $\mathcal{L} = -\frac{m_0}{az+1}\sqrt{1-\dot{z}^2}$
- I have also proven $z(t) = \frac{\sqrt{a^2 t^2 + 1} - 1}{a}$ is unique to the EOM associated with $\mathcal{L}$

- Reviewing dark matter problem
- Reviewing orbital motion $\mathcal{L} = \mathcal{L}_{CM} + \mathcal{L}_{rel}$

End